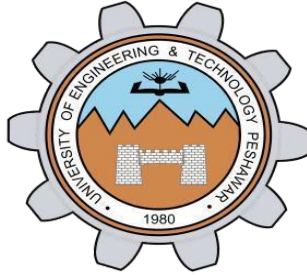


VISION TRANSFORMER DRIVEN MANGO LEAF DISEASE DETECTION WITH TARGETED SPRAYING USING UAV



SESSION: 2022-2026

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A Thesis
Submitted to the UET Peshawar as a
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PESHAWAR, PAKISTAN

DECLARATION

We declare that this dissertation is our original work, undertaken with integrity and in accordance with the guidelines provided by Department of Computer Systems Engineering, University of Engineering & Technology, Peshawar. All sources used in this dissertation are appropriately acknowledged and referenced. Any assistance in preparing this dissertation is duly acknowledged.

By signing below, we affirm the originality of our thesis.

DEDICATION

Every challenging project requires dedication and the guidance of elders who are close to our hearts. This work is lovingly dedicated to our beloved parents, whose affection, support, and continuous prayers made this achievement possible, and to our respected teachers for their guidance and inspiration.

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Abstract

Mango is one of the most economically important fruit crops in Pakistan and plays a significant role in the agricultural sector and export market. However, mango production is severely affected by various leaf diseases, including Anthracnose, Powdery Mildew, Bacterial Canker, Dieback, Gall Midge, and Sooty Mould. These diseases reduce crop quality, decrease yield, and cause substantial economic losses for farmers and exporters. Traditional disease monitoring methods rely heavily on manual inspection and blanket pesticide spraying, which are time-consuming, labor-intensive, costly, and often ineffective in large orchards. Therefore, the development of an intelligent and automated disease detection system is essential for improving disease management and promoting sustainable agricultural practices. This project presents a Vision Transformer (ViT)-driven mango leaf disease detection and targeted spraying framework using an Unmanned Aerial Vehicle (UAV). The proposed system combines advanced deep learning techniques with precision agriculture to provide automated disease identification and localized pesticide application. A publicly available mango leaf disease dataset consisting of multiple disease classes was utilized for model development. To improve the robustness and real-world applicability of the system, extensive data augmentation techniques were applied, and CycleGAN-based domain adaptation was employed to generate drone-style synthetic images that closely resemble aerial imagery captured during UAV operations. This approach helped bridge the gap between laboratory datasets and real-world field conditions.

For disease classification, a DeiT-Small (Data-efficient Image Transformer) architecture was adopted due to its ability to capture global contextual information through self-attention mechanisms. Unlike conventional Convolutional Neural Network (CNN) approaches, which primarily focus on local image features, the Vision Transformer effectively learns long-range relationships between image regions, enabling more accurate disease recognition under varying environmental conditions. A two-phase training strategy, custom LayerNorm classification head, and multiple regularization techniques were employed to improve model generalization and reduce overfitting.

Experimental evaluation demonstrated that the proposed model achieved excellent classification performance across all disease categories. The trained Vision Transformer achieved approximately 97% validation accuracy while maintaining high precision, recall, and F1-score values on both real and synthetic test datasets. Comparative analysis indicated that the proposed ViT-based framework outperformed many conventional CNN-based approaches reported in the literature by providing improved feature representation, stronger generalization capability, and enhanced robustness against variations in lighting, background clutter, and image perspective. The CycleGAN-based domain adaptation strategy further improved deployment readiness by enabling the model to perform effectively on drone-style imagery.

To support practical implementation, a custom Hexacopter UAV equipped with an imaging system and targeted spraying mechanism was designed and fabricated. During operation, the UAV captures images of mango trees, identifies diseased regions using the trained Vision Transformer model, and generates disease location information for selective pesticide application. This targeted spraying approach significantly reduces unnecessary pesticide usage, lowers operational costs, minimizes environmental contamination, and improves overall crop management efficiency.

The developed system demonstrates the effectiveness of integrating Vision Transformer technology with UAV-based precision agriculture for intelligent disease monitoring and intervention. The proposed framework provides a scalable, cost-effective, and environmentally sustainable solution for mango orchard management and has the potential to be extended to other crops and agricultural disease detection applications in the future.

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LIST OF ABBREVIATIONs

Abbreviation	Definition
AI	Artificial Intelligence
ANN	Artificial Neural Network(s)
BPNN	Back Propagation Neural Network (Multilayered)
CCA	Canonical Correlation Analysis
CNN	Convolutional Neural Network(s)
DL	Deep Learning
DCNN	Deep Convolutional Neural Network
YOLOv5	You Only Look Once version 5 (CNN model)
YOLOv3	You Only Look Once version 3 (CNN model)
UAV	Unmanned Aerial Vehicle (drone)
ViT	Vision Transformer

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Project title: Vision transformer driven mango leaf disease detection with targeted spraying using UAV

CHAPTER 1 INTRODUCTION

Introduction

Agriculture is considered to be the backbone of developing nations. But, as rapid growth in population leads to increasing the requirement for food and management systems, technology has emerged as a pro-active solution. Automated detection systems for disease in crop and ensuring crop productivity is one such technology required to protect high-value crops like *Mangifera indica* from damage. The need for technology for precision agriculture is driven by a desire to make agricultural management easier by utilizing advanced technology [14].

Diseases such as Anthracnose, Powdery Mildew and Bacterial Canker can have a great impact on reducing crop production. Conventional methods for dealing with such diseases include manual examination and spray techniques wherein chemicals are applied to the fields indiscriminately. Both these techniques are quite expensive and inefficient and may lead to unnecessary use of chemicals with detrimental effects on the environment [9], [10].

Despite the technological advances, some limitations persist in current disease detection systems. Most approaches use manual visual inspection or traditional Convolutional Neural Networks (CNN) techniques which might fail to detect certain complex diseases accurately due to changes in their symptoms [10], [12]. In addition to that, current systems lack the ability to respond instantly in cases of disease occurrence. Instead, the delay between detection and application of pesticides might cause damages and losses [16].

In an attempt to tackle these problems, this study explores the application of Vision Transformer (ViT) models for the detection of mango leaves diseases in combination with an unmanned aerial vehicle (UAV)-assisted targeted spraying system. In contrast to traditional CNN methods, Vision Transformers incorporate self-attention layers enabling the network to perceive both local and global image data efficiently [1], [2]. The proposed method was evaluated using an extensive dataset of mango leaf images.

Moreover, a specialized UAV called Hexacopter, which incorporates a mechanism to target spray, has been added to the system. This allows the use of pesticides to target only the affected area, which will reduce unnecessary use of chemicals. The process of "blanket spraying," however, results in unnecessary pesticide application, leading to higher costs for farmers, and poses a danger to both humans and the environment [9]. Even though there exist some forms of automated systems, they still rely on out-of-date CNN architecture that lacks global understanding, [12] as well as having no hardware for quick action [16]. It is absolutely vital to incorporate a fully developed vision transformer along with a UAV for pesticide targeting [1]. This will prove crucial for ensuring healthy crops and minimizing pesticide waste, which will increase profitability of mango plantations.

Problem Statement

While modern agricultural disease management techniques are effective, there are still several shortcomings associated with the efficiency of the current state-of-the-art disease detection and treatment process [10]. Specifically, manual field inspection is extensively used today, although this process is quite resource-intensive, time-consuming, and can be highly dependent on human errors [10]. Besides, the current practice of blanketing spraying includes uniform distribution of pesticides on the crops without regard to the presence of infections in plants, leading to unnecessary use of chemicals and associated additional costs [9].

Despite the variety of automated disease detection systems currently available, most of them are designed using the traditional CNN models and may fail to accurately recognize diseases and extract long-range connections between features of diseases [12]. In addition, existing automated systems are mainly focused on the recognition stage, while little attention is paid to the subsequent implementation of the treatment stage on-site [16]. As a result, the delay in disease detection and its treatment can increase infection spread and additional crop losses.

In order to overcome the mentioned drawbacks, this project focuses on developing a novel disease detection technique combining a Vision Transformer and UAV-based targeting spraying system [1], [2].

1.1 Background

1.1.1 Evolution of Agricultural Monitoring Systems

There has been significant development in the ways agriculture is monitored over time. In the past, manual monitoring of crops involved conducting inspections on a field level by farmers and agricultural extension personnel. Despite being the most common practice, it took a lot of time and effort with human factors affecting its efficiency [10].

Satellite imaging and sensors for field monitoring became a step forward as they allowed larger fields to be observed from above. These technologies still failed to detect leaf diseases in their earliest stage, leading to late identification of any plant infections. Disease symptoms might not be noticed until the disease spread over the majority of the orchard. Therefore, fungal infections would only be seen when it is too late since the crop will be significantly damaged [10].

1.1.2 Vision Transformer Approach in Disease Detection

The progress in the Deep Learning field has played a pivotal role in the emergence of automated disease detection systems. Specifically, CNNs have become popular for their ability to perform image-based disease detection. However, Vision Transformers (ViTs), based on the use of self-attention, are emerging as another approach that provides better results in visual data analysis [2]. Whereas the CNN model relies primarily on the local analysis of image data, ViTs are able to establish dependencies throughout the whole image, thus helping to detect diseases spread all across the leaf [12].

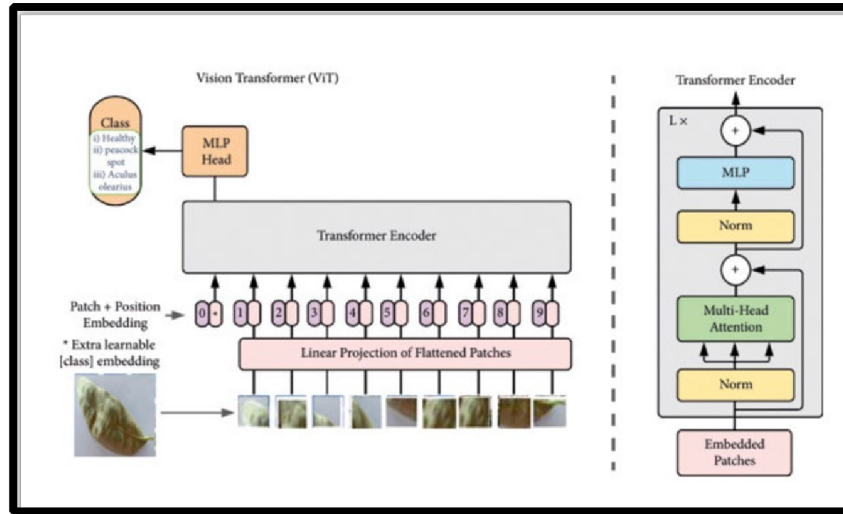


Figure 1 Vision Transformer (ViT) architecture adopted for mango leaf disease classification from leaf images.

1.1.3 Challenges of Existing Systems

Despite all these advancements, however, there are still several shortcomings in the existing agricultural monitoring systems. A considerable number of existing solutions are software-focused, addressing solely the detection part of the process and lacking a means of taking action based on the results, namely applying sprays [16]. Furthermore, traditional drones used in agriculture tend to operate manually, leaving insufficient room for immediate decisions, thus requiring constant human presence throughout the monitoring and treatment process.

Perhaps another crucial problem lies in the fact that the connection between an intelligent disease detection algorithm and the corresponding hardware system tends to be underdeveloped. This inevitably makes the idea of precision farming impossible to achieve because these two stages are considered separate parts of the process.

1.1.4 UAV Integration for Precision Spraying

UAV technology, mainly involving multirotor hexacopters, has drawn much attention regarding its applicability in precision agriculture due to VTOL capability, stable hovering, and payload carriage capabilities that enable UAV usage in close-up crop surveillance and selective spraying applications [16]. In mango trees, UAVs can be used to inspect leaves and selectively spray the pesticide only when necessary. Several studies confirm that targeted spraying reduces pesticide usage by 30%–50% in comparison with the traditional blanket spraying technique [9].

Nevertheless, there are some major issues regarding UAVs used in disease management applications:

The Lack of ML Integration Capability: Many current approaches fail to connect disease detection models directly with the spraying system preventing them from executing an instant action after disease detection [16].

- Computational Capabilities: Complex deep learning algorithms usually require a great amount of computational power making it difficult to deploy them on-board of UAVs.
- Lack of Data: Due to the absence of publicly available aerial data including pictures of mango leaves taken by a UAV camera.

A more recent trend in research is aimed at solving the aforementioned problems through the development of disease detection systems using ViT models integrated with UAVs. The usage of multispectral images for finding early signs of plant diseases even before visual symptoms can be seen has been researched as well. This combination allows for disease detection and application of pesticides in one process, providing efficiency in agricultural work [16].

1.2 Motivation

Mango tree diseases pose a serious threat to food security and the agrarian economy [10]. The key problem for modern agronomy lies in finding effective solutions to deal with these issues. Regrettably, the basis for traditional surveillance is reactive approaches which cause delays and amplify the consequences of infections. Limited budget and increased crop losses occur due to these delays. Such constraints become increasingly ineffective under increasing production costs and environmental regulations.

The use of UAVs and ViT algorithms represents a new approach to overcome these constraints. By implementing these technologies, farmers will be able to switch from reactive to proactive approaches. It becomes important for farmers who want to increase the quality of their products and reduce the need for pesticides [9]. In addition, the system's flexibility will help it adapt to various orchard configurations..

1.2.1 The Increasing Need for Precision Agriculture

With the increasing number of people on earth, there is increasing need for enhanced agricultural output and better yield from crops. Although traditional forms of agriculture remain significant, they can sometimes face problems such as labor shortage, disease outbreak, and management of resources. However, precision agriculture can serve as a viable solution to all these problems since it enables farmers to monitor and control their crops in a more precise manner. In this context, targeted spraying of crops will help save chemicals from wastage and cost reduction [9]. Without such measures, farmers will find themselves facing lower yields, increased costs of production, and even environmental harm.

1.2.2 Limitations of Existing Technologies

The most used conventional technology is the application of the required pesticides from spray machines mounted on tractors or manual spraying without paying attention to the location of infected trees [9]. It leads to overexposure of chemicals to the environment and excessive exposure of

field workers. Furthermore, it is difficult to perform disease detection accurately under natural conditions due to the influence of such factors as varying lighting and background characteristics, as well as different leaf shapes. Conventional convolutional neural networks have some limitations in recognizing complex diseases in a natural environment [10].

1.2.3 Advancements in Deep Learning and Robotics

In recent years, significant progress has been made in the area of deep learning and robotics that have become promising in terms of automation. Improvements in UAVs and flight controller algorithms resulted in stable work during automatic flights in agriculture. On the other hand, there was progress in deep learning with Vision Transformers that proved their effectiveness in image recognition tasks by capturing local and global features of an image [1], [2].

1.2.4 Integration with UAV Frameworks

The integration of machine learning-based disease detection with a hexacopter UAV platform provides a practical approach to automated crop monitoring and treatment. By utilizing onboard cameras, sensors, and control systems, the platform can support image acquisition, data processing, and targeted field operations within a single framework. This combination enables disease detection results to be linked directly with spraying actions, improving the overall efficiency of orchard management.

1.2.5 Societal and Economic Advantages

Use of such a technology will help in ensuring various social and economic advantages as well as benefits for the environment. Application of pesticides where they are required will help to decrease the quantity of chemicals used in agriculture and cut down costs [9]. Moreover, increased crop health and the quality of the fruits will positively affect their value and availability in the foreign markets.

1.2.6 Contribution to Sustainable Agriculture

It is worth noting that the use of vision transformers in agriculture will also help to promote sustainable development through the efficient use of pesticides. Targeted spraying will prevent additional applications and the consequent contamination of the soil and other nearby areas [9].

1.3 Project Goals

The purpose of the current project is to design a system that uses Vision Transformers for disease detection and provides for UAV spraying. The technology is supposed to ensure timely disease detection as well as efficient use of pesticides. With the help of a hexacopter, it will be possible to monitor crops and apply pesticides when it is required.

1.3.1 Develop a ViT Detection Model

The primary objective of this study would be the development of a detection model for mango leaf diseases like anthracnose and powdery mildew. This model will be designed based on the Vision Transformer (ViT) architecture and will be trained using the dataset collected for the purpose. Attention techniques will be employed for the better identification of diseases under various light conditions.

1.3.2 Design and Fabricate a Hexacopter UAV

The second objective of the proposed work includes the fabrication of a hexacopter UAV which will serve as the base for the detection system along with other components. Hexacopters have proven their efficacy to be a more reliable alternative for the detection of diseases and spraying operations than the commonly used quadcopters because of their high payload and stability.

1.3.3 Develop Targeted Spraying System

As a third objective, a spraying mechanism will be developed and incorporated into the UAV to spray pesticides at specific locations after disease recognition. The pump and relay-based control system will be used for this purpose.

1.3.3 Enhancing Real-Time Efficiency

A third objective of the project would be to minimize the gap between disease diagnosis and pest control through increased efficiency in image processing and real-time communication between components.

1.4 Scope and Assumptions

In terms of scope, the research would cover the design and development of a disease detection and pesticide application system for use in mango orchards. The proposed system incorporates the use of vision transformers in the recognition of common mango leaf diseases such as anthracnose, bacterial canker, and powdery mildew. Other aspects within the scope include the design and construction of the hexacopter drone as well as the spraying device.

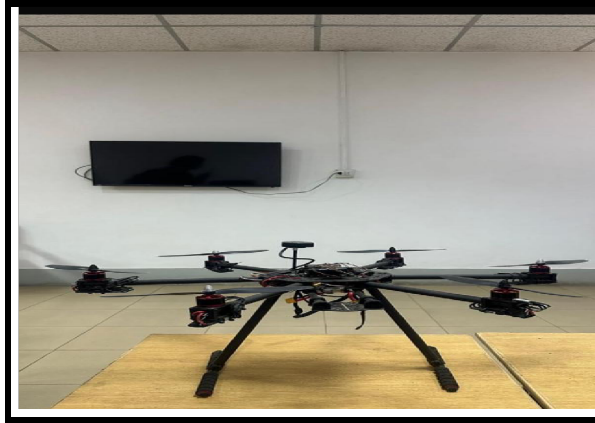
The successful design and development of the proposed system are contingent on access to a labeled mango leaf diseases data set as well as other UAV hardware such as controllers, ESCs, motors, etc. Further, the field tests will be done under favorable climatic conditions.

1.4.1 Identification of Specific Diseases

For the purposes of this research, particular attention is being paid to bacterial and fungal mango leaf diseases commonly found in South Asia due to their harmful effects on crop yields. The vision transformer-based approaches for classification are used in this paper [1], [2].

1.4.2 Hardware Architecture (Hexacopter)

A hexacopter configuration is selected to provide improved flight stability and operational reliability. The hardware scope includes the airframe, propulsion system, power distribution components, flight controller, and the spraying payload required for targeted pesticide application.



Hexacopter

Figure 2 Hexacopter UAV architecture used for targeted spraying of mango trees based on detected leaf diseases.

1.4.2 Automatic Spraying Protocols

Automatic control of the spraying operation is possible due to the decision-making algorithm of the system. Upon detecting a disease, the system can trigger the spraying mechanism to spray the diseased area. This method allows for efficient use of pesticides without overusing them [9].

1.4.3 Data and Resource Requirements

It is presumed that the development of the proposed system would require a labeled mango leaf disease dataset and computing power to train the Vision Transformer model. Moreover, mechanical and electronic hardware required for building the UAV will be presumed to be available. The resources are necessary for the proper functioning of the system.

1.5 Importance of the Study

The current research adds to the body of knowledge regarding the development of intelligent agricultural disease management systems using UAVs. By applying machine learning algorithms in tandem with UAV-assisted targeted pesticide application, the limitations of traditional disease management techniques are being addressed.

1.5.1 Improved Crop Productivity and Health

With the help of early detection and treatment capabilities offered by the system, one might be able to decrease the propagation of diseases among trees. Using Vision Transformers for disease classification can increase chances that diseased plants will be recognized and properly treated in a timely fashion [1], [2]. Consequently, healthier crops and increased yield can be expected.

1.5.2 Advances in Agri-Robotics

This work showcases the application of Vision Transformer for detecting diseases combined with the use of unmanned aerial vehicles for agricultural purposes. Such integration can illustrate the possibility of applying artificial intelligence in modern agriculture [16].

1.5.3 Sustainability

Applying pesticides specifically to diseased areas might help save both the environment and additional money for the farmer since unnecessary chemicals would not be sprayed on unaffected plants [9]. Lessening usage of inputs can also lead to better economical results.

1.6 Structure of the Thesis

This work is divided into several chapters which cover different aspects of research and systems development.

1.6.1 Introduction

In the first chapter, the problem is introduced and its background is discussed briefly. In addition to this, problem statement, objectives, scope of this research, assumptions, and significance of the project are also covered in this chapter.

1.6.2 Literature Review

Previous researches and technologies on the areas of plant diseases diagnosis, Vision Transformers, CNNs, and UAVs' utilization in agricultural fields have been reviewed in this chapter. Research gaps associated with the current study have been identified in this chapter as well [10], [12].

1.6.2 Methodology

In the third chapter, the methodology utilized for conducting this research work has been described. It covers fabrication process of the hexacopter platform, hardware configuration, spray nozzle development, dataset construction, and vision transformer model implementation.

1.6.3 Results and Discussion

Experimental results and analysis of the developed system are provided in this chapter. This chapter includes results of disease diagnosis task, analysis of UAV performances, spray system analysis, and discussion of the results.

1.6.4 Conclusion and Future Work

This concluding chapter highlights key findings and contributions made through the course of the research work. Limitations associated with the current work have also been mentioned in this chapter.

Chapter 2 Literature Review

2.1 Precision Agriculture and Diseases Affecting Mango Leaves

The mango plant (*Mangifera indica*) is considered one of the popularly planted fruit crops that have significant value to the agricultural economy in many countries. Unfortunately, mangoes are susceptible to several diseases including anthracnose, powdery mildew, and bacterial canker, which are capable of affecting fruit quality and yield [2] [5]. To effectively prevent and control the impact of diseases on mango fruit, disease detection needs to be performed accurately and quickly, otherwise, the infections will spread quickly resulting in crop losses [4] [6].

Traditionally, disease detection on mango plantation farms has been conducted through manual methods. Despite the popularity of manual disease detection techniques, these methods may be cumbersome, laborious, and experience-based [10]. The problems associated with traditional disease detection methods have motivated the use of digital solutions to monitor crop fields.

The recent developments in Machine Learning (ML) and Deep Learning (DL) have offered new possibilities for automating the process of disease identification in plants [14]. The application of image detection techniques in the agricultural industry has become widespread due to its capability of assisting the diagnosis of diseases at a much faster pace. This chapter focuses on the history of the development of computer vision techniques that have been used in agriculture, starting from traditional ML models up until Convolutional Neural Networks (CNN) and Vision Transformer models (ViT). Moreover, the chapter discusses the role of UAVs in precision agriculture and addresses the problem of this study.

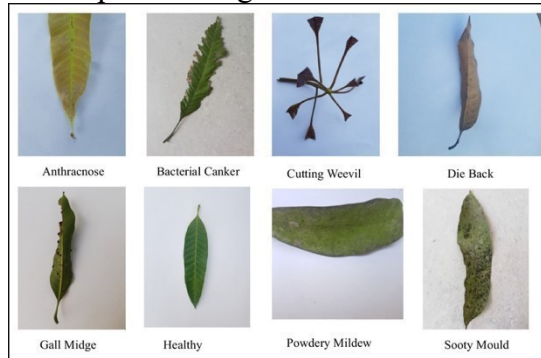


Figure 3 Example mango leaf samples from eight disease categories: (a) Anthracnose, (b) Bacterial Canker, (c) Cutting, (d) Weevil, (e) Dieback, (f) Gall Midge, (g) Healthy, (h) Powdery Mildew, and (i) Sooty Mould.

2.1.1 The Effects of Mango Leaf Diseases on Economics

The effect of mango diseases is widespread and includes different parts of the plant, such as leaves, flowers, stems, and fruits, leading to poor crop yield and quality [10]. Anthracnose, which is caused by the pathogen *Colletotrichum gloeosporioides*, is among the most destructive fungal plant pathogens in the context of mango production [6]. In addition, there is powdery mildew disease that is able to affect orchard productivity. It should be noted that a significant percentage of unprotected mango plants suffer from such disease [10]. The economic effect makes it clear

that the detection and identification of the plant diseases play a key role in the development of an automated detection system [4].

2.2 Classic Approach for Machine Learning Techniques

Initially, research in the field of plant disease automatic detection involved extracting manually engineered features using classic machine learning approaches.

2.2.1 Techniques and Results

Such techniques often began with pre-processing of images through techniques like segmentation and enhancement to identify hand crafted features like textures with the use of techniques like Gray Level Co-occurrence Matrix (GLCM) and Local Binary Patterns (LBP) [4], [16]. After this, the features thus identified were subjected to classifiers like Support Vector Machines (SVM), K-Nearest Neighbors (KNN), and Artificial Neural Networks (ANN) [7].

Classification accuracies have been found in the range of 80%-95% with controlled data [7], [10]. Multi-class classification techniques based on diseases had comparatively lower accuracy levels of 66% in the case of KNN and 62% in the case of ANN [7].

2.2.2 Limitations of Classical Approaches

While classical machine learning techniques produced positive outcomes in a controlled environment, some disadvantages hindered the use of such methods [4]:

- **Dependence on Environmental Conditions:** Features selected manually could be sensitive to changes in lighting conditions and shadows that usually existed within an orchard environment [16].
- **Bad Generalization Capability:** Poor generalization capability was also observed among models when trained using data from one set of images but evaluated using different data [10].
- **Feature-Based Approach:** Performance of the proposed system relied significantly on selecting proper features of an image, which required considerable domain knowledge [4].

2.3 Deep Learning: CNN for Disease Classification

The advent of Deep Learning, specifically, the CNN model, played an important role in the advancement of image-based disease detection through the automated feature extraction process [10].

2.3.1 Adoption of Architecture and Performance Benchmarking

The use of transfer learning (TL) that incorporates pre-trained architectures including the ResNet and EfficientNet, is common practice in agricultural image classification due to its benefits on model performance and training processes [5], [6].

- **Higher Accuracy:** Multiple CNN models showed classification accuracies that exceeded 99% on the disease database. ResNet models gave an accuracy of 99.88%, while customized models showed an accuracy of 99.55% on the MangoLeafBD data set [6], [13].
- **Efficiency:** Models were optimized through techniques like Dynamic Quantization (DQ) to minimize computational cost [6].

2.3.2 Key Weaknesses of CNNs

Despite being highly effective in most cases, CNN-based approaches face some drawbacks [12]:

- **Small Context Window:** CNNs pay attention mainly to local parts of an image and fail to take into account long-range relations throughout an image. In such cases, it becomes difficult to detect small objects or disease symptoms scattered throughout an image [12].
- **Performance Reliability Under Field Conditions:** Their accuracy reduces when used with orchard images with different lightening, shadows, and overlapped leaves [10].

2.4 Vision Transformers (ViTs)

A recent innovation in the field of image recognition is the development of vision transformers that rely on the concept of self-attention, which enables the extraction of relations between parts throughout the entire image [12], [15].

Vision Transformers divide the input images into patches and apply self-attention operations based on Transformers. As such, the model can detect interactions between various image sections, enabling learning of global contextual cues [12], [15]. The features are very handy in detecting small signs of diseases that might appear in several regions of a leaf.

- **Good Classification Accuracy:** Recent research has confirmed that Vision Transformers achieve very high classification accuracy close to 100% when applied to the MangoLeafBD data set [1], [15].
- **Capacity to Detect Multiple Classes:** Vision Transformers have been found capable of classifying multiple mango leaf diseases in one architecture [3].

2.4.2 The Deployment Challenge for ViTs

Despite evidence that Vision Transformer models are extremely effective in classifying plant diseases [1], there is a significant issue with implementing such approaches. Most current literature focuses on images collected from a close range within controlled environments. Far fewer studies have been conducted in relation to integrating ViTs with UAVs as well as real-time disease management tools. These issues underscore the necessity for future studies into ViT-powered solutions for UAV-based disease detection and management systems [1].

Vision Transformer

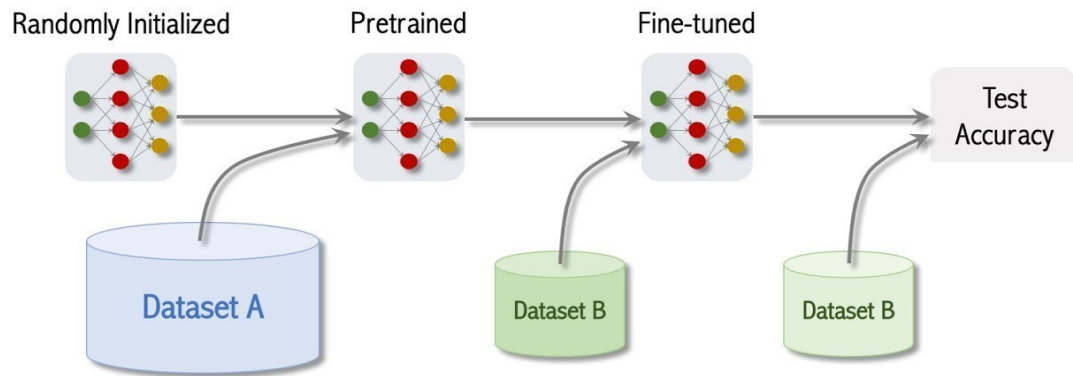


Figure 4 Vision Transformer training pipeline showing pretraining on Dataset A and fine-tuning on Dataset B before evaluation. Image adapted from an online source.

2.6 Research Gap and Project Relevance

Based on the literature discussed in this chapter, much progress has been observed in the development of disease detection algorithms as well as UAV technology used in agricultural applications. Vision Transformer models show great potential in performing plant disease classification, whereas agricultural UAVs have been extensively applied for crop monitoring and pesticide spraying purposes. Nevertheless, few studies have been conducted on incorporating both technologies into one framework to perform disease detection and treatment.

This study tries to fill the existing gap by providing the following innovations:

1. **Integrated ViT-Hexacopter Framework:** The current research integrates a disease detection algorithm based on the Vision Transformer with the hexacopter UAV for monitoring and spraying pesticides on mango tree plants.
2. **Onboard Computing System Design:** This study explores the integration of the plant disease detection algorithm into the computing framework of the UAV for image processing purposes.
3. **Mango Leaf Diseases Detection:** The proposed system will be able to detect and classify multiple types of leaf diseases in mango plants.

CHAPTER 3 METHODOLOGY

3.1 Research Methodology Overview

In this chapter, the methodology which was adopted for designing the automatic detection of disease of mango leaves using UAV in the context of precision agriculture will be discussed. The methodology consists of the following five steps: Dataset collection and preparation; image synthesis via CycleGAN; vision transformer-based classification training and testing of model on synthesized as well as original datasets; and practical evaluation of the model's performance in real life applications.

One of the primary issues dealt with in this research is the problem of misalignment between the ground level images of mango leaves used in publicly available datasets and the aerial images captured from UAVs [9], [15]. Most publicly available datasets comprise of ground level images of plants, therefore the need for domain adaptation in this context became inevitable. For this purpose, image augmentation and image style transfer methods based on CycleGAN have been used for generating artificial images resembling those obtained from UAV [12].

The proposed framework can be implemented using a monitoring system based on UAV. In practice, the UAV will fly through the orchard and take pictures of mango trees. These images will then be analyzed using the disease classification algorithm. This way, infected areas can be saved along with GPS coordinates. And using these coordinates, the pesticide can be applied only to those places, where the disease is detected, but not to the entire orchard [16].

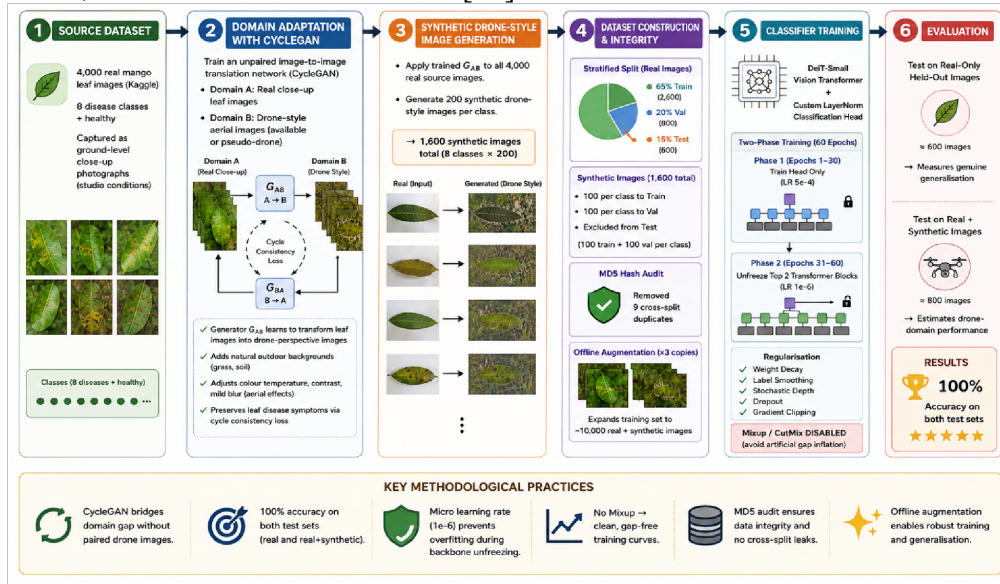


Figure 5 Methodology Pipeline.

3.2 System Architecture

The system designed involves two major phases: an offline training phase and an online inference phase. Offline phase is the process of data preparation and model training, while the online phase is the process of disease classification in the system run.



Figure 6 Fabricated Hexacopter UAV with integrated camera and targeted spraying mechanism, assembled during the hardware development phase.

3.2.1 Off-Line Training Pipeline

Before the model's deployment, the off-line training pipeline was used to prepare the dataset and train the model for classifying diseases in the crop. These were the essential steps involved in this stage:

- **Dataset Preparation:** The mango leaf disease dataset with 4000 images collected from Kaggle was stratified into training, validation and test datasets in the proportions of 65%, 20% and 15% respectively. Stratified sampling ensured that each category of diseases appeared proportionately across the three subsets [13].
- **CycleGAN Training:** In order to generate the common visual features from the UAV images, a CycleGAN model was trained on mango leaves images and images from previous drone operations.
- **Datasets merging and duplicate removal:** Real and synthetic images are going to be merged into a larger dataset. In order to make sure that there weren't the same images in training and test datasets, a duplicate removal was done.
- **Data Augmentation:** The complex data augmentation technique was used for the overall dataset to improve image diversity and help the model generalize better. This provided about 18,000 training images and 4,800 validation images to train the model.
- **Model training:** DeiT-Small Vision Transformer [2] with custom LayerNorm classifier head. The two-stage progressive unfreezing training approach was used (Section 3.5).

3.2.2 Online Inference Pipeline

While the system is running, the UAV captures geo-tagged RGB images taken from an altitude of 10 to 30 metres above the canopy level of the orchard. The captured images are sent to the ground station where they are converted into 224x224 pixel images and normalized according to the ImageNet mean values for the detection process using the pre-trained DeiT-Small architecture [5]. Disease symptoms are identified, marked on the images, and their GPS coordinates are noted. Based on the marked disease areas, a treatment map is made to assist the UAV in applying pesticides. This strategy helps the application of pesticides only on areas affected by the diseases [16].

3.3 CycleGAN for Domain Adaptation

The Cycle Consistent Generative Adversarial Network (CycleGAN) was chosen to be used as the domain adaptation method due to its ability to learn image-to-image translations even when using unpaired data sets. This is especially important for the current study since there is no data set that contains mango leaves imaged both at ground level and at UAV heights [9], [15]. This method was first introduced in a paper written by Zhu et al. (2017).

3.3.1 Network Architecture

The CycleGAN architecture is built using four neural networks:

- Generator G_{AB} (Leaf $\rightarrow$ Drone): ResNet-6 generator employed for generating drone-style images from close-up mango leaves images. This neural network has six residual blocks between the encoder and decoder, enabling image translation as well as manageable computation costs.
- Generator G_{BA} (Drone $\rightarrow$ Leaf): A similar architecture with ResNet-6 generator, generating leaf-style images from drone images. It is employed for implementing the cycle consistency principle.
- Discriminator D_A : A PatchGAN discriminator evaluating 70×70 patches and classifying input images as either belonging to the real close-up leaf domain or as synthetic.
- Discriminator D_B : PatchGAN discriminator differentiating between the real drone images and synthetic drone-style images generated by Generator G_{AB} .

PatchGAN was chosen over full-image discrimination due to local realism in texture synthesis encouraged by patch-level feedback [12].

3.3.2 Loss Functions

CycleGAN training involves a combination of three losses:

Adversarial Loss: GAN Loss based on the mean squared error in line with LSGAN formulation that is less instable in training than the binary cross-entropy function.

Cycle Consistency Loss: Imposes constraint on the round-trip reconstruction, preventing mode collapsing and retaining information on the presence of diseases in the translated images.

Identity Loss: Penalties the generator for making any changes to images in the target domain, ensuring stable colour maps within both domains.

The overall loss is: $L_{total} = L_{GAN} + \lambda_{cyc} \times L_{cyc} + \lambda_{id} \times L_{id}$, where $\lambda_{cyc} = 10$ and $\lambda_{id} = 5$, based on the values used in the original CycleGAN article.

3.3.3 Training Parameters

CycleGAN was trained for 25 epochs, Adam optimiser with $\beta_1=0.5$, $\beta_2 = 0.999$, learning rate of 2×10^{-4} , and a batch size of 4 limited by VRAM of P100 GPU. The linear decay learning rate schedule was introduced starting from epoch 13. Replay buffer with 50 images for each discriminator was maintained during the adversarial training. The hyperparameters are listed in Table 3.1.

Hyperparameter	Value	Rationale
Epochs	25	Sufficient for loss convergence; G loss declined from 6.26 to 3.44
Learning rate	2×10^{-4}	Standard CycleGAN recommendation
Batch size	4	VRAM-constrained (P100 16 GB)
Lambda cycle (λ_{cyc})	10	Balances cycle consistency
Lambda identity (λ_{id})	5	Preserves colour mapping stability
Replay buffer	50 images	Stabilises discriminator training
Generator	ResNet-6	Efficient; suitable for 224×224 input
Discriminator	PatchGAN 70×70	Local texture realism

Table 3.1 — CycleGAN Training Hyperparameters

3.4 Vision Transformer Model Architecture

The classification model used here is the DeiT-Small (Data-efficient Image Transformer) introduced by Touvron et al. (2021) [2]. The DeiT-Small was chosen instead of other ViT architectures for a few compelling reasons based on the literature on efficient vision transformers for agriculture [1], [12]:

- Parameter efficiency: DeiT-Small uses around 22 million parameters such that it is feasible to train on an augmented dataset of around 18,000 images without overfitting [2].
- Data efficiency: Contrary to the vanilla ViT, the DeiT uses knowledge distillation during training such that it can achieve good ImageNet results without using the large-scale JFT-300M dataset [2].
- Compatibility: DeiT-Small supports 224×224 pixel input images using 16×16 patches with 196 patch tokens and one classification (CLS) token using 384 dimensions per token [1].
- Pre-trained quality: Pre-trained weights on ImageNet-1k are publicly available through timm and give excellent visual feature representations for downstream fine-tuning [5], [6].

3.4.1 Standard DeiT-Small Architecture

The DeiT-Small backbone network consists of 12 encoder blocks, each block containing MHSA along with a two-layer MLP structure. This is connected through residual connections while also using LayerNorm [2] in between the blocks. As the architecture utilizes self-attention mechanisms, this allows for the learning of long-range spatial relationships within leaf images, which have been observed as an important feature for disease detection in earlier stages [12].

Parameter	Value
Number of transformer blocks	12
Embedding dimension	384
Number of attention heads	6
MLP hidden dimension ratio	4× (1,536)
Patch size	16×16 pixels
Input resolution	224×224 pixels
Total tokens	196 patch tokens + 1 CLS token
Total parameters	~22 million
Pre-training dataset	ImageNet-1k
Internal normalisation	LayerNorm (per block)

Table 3.2 — DeiT-Small Architecture Parameters

3.4.2 Custom LayerNorm Classification Head

DeiT’s default classification head, i.e., a single linear projection from the CLS token, was replaced by a three-layer MLP using LayerNorm normalisation at each hidden layer. The following three design motivations justify the selection of this head over BatchNorm heads [2], [12]:

- LayerNorm normalises each sample independently along the feature dimension, making it consistent with DeiT’s overall architecture where each transformer block performs LayerNorm [2].
- BatchNorm calculates statistics along the batch axis and keeps running estimations that might lead to differences between training and inference batches [10], causing a train/test disparity.
- LayerNorm is not affected by the batch size, providing the same normalisation at training, validation, and test time, which is important when deploying DeiT on UAV onboard hardware with varying batch sizes [16].

Layer	Operation	Output Dim	Purpose
0	LayerNorm(384)	384	Normalise CLS token
1	Linear(384→256)	256	Feature compression
2	LayerNorm(256)	256	Hidden normalisation
3	GELU()	256	Non-linear activation
4	Dropout(p=0.25)	256	Regularisation
5	Linear(256→128)	128	Second compression
6	LayerNorm(128)	128	Hidden normalisation
7	GELU()	128	Non-linear activation
8	Dropout(p=0.15)	128	Regularisation
9	Linear(128→8)	8	Class logits output

Table 3.3 — Custom LayerNorm Classification Head Architecture

3.5 Two-Phase Progressive Training Strategy

A two-stage progressive unfreezing strategy was employed to prevent the catastrophic forgetting of ImageNet pre-trained features whilst allowing adaptation specific to the mango disease dataset [5], [6]. Initial experiments showed that applying full model finetuning immediately at a fixed learning rate led to quick overfitting, an experience echoed in transfer learning research in small agricultural data sets [11].

3.5.1 Stage 1 – Classification Head Training (Epochs 1–30)

In Stage 1, all parameters of the DeiT backbone architecture were frozen, and only the LayerNorm classification head trained. This ensured the randomised classifier had sufficient time to train on its own and learn to project from the pre-trained feature space onto the eight mango diseases classes without interfering with the ImageNet pre-trained representations [2], [6]. A learning rate of 5×10^{-4} , weight decay of 0.05 and cosine annealing learning rate scheduler after a five-epoch linear warmup period was used.

3.5.2 Phase 2 Selective Backbone Unfreezing (Epochs 31–60)

During Phase 2, the two topmost blocks (blocks 10 and 11) of the transformer were unfrozen and tuned using the micro learning rate of 1×10^{-6} — 500 times less than that for the head learning rate. The method enabled the top blocks that encoded high-level features related to the tasks [12] to gradually adjust towards classifying mango diseases without disturbing low-level visual information. Simultaneously, the weight decay was doubled to 0.10 as the number of trainable parameters was increased from about 0.5M to 3.7M.

Parameter	Phase 1 (Ep 1–30)	Phase 2 (Ep 31–60)
Trainable modules	Head only	Head + top 2 blocks
Trainable parameters	~0.5M	~3.7M
Head learning rate	5×10^{-4}	5×10^{-4} (cosine decay)
Backbone LR	N/A (frozen)	1×10^{-6} (micro)
Weight decay	0.05	0.10
Optimiser	AdamW	AdamW (extended)
LR schedule	Cosine + 5ep warmup	Cosine (continued)
Mixup/CutMix	Disabled	Disabled
Observed gap	< 3%	< 3%

Table 3.4 — Two-Phase Training Configuration

One of the important diagnostics performed was the effect of Mixup and CutMix augmentations. Even though these were noted to be helpful for training ViT [3], in this setting they resulted in a false training vs. validation accuracy spread by 10 -13 %. This happened because Mixup's soft-labeling mechanism lowers the observed training accuracy, whereas validation is evaluated using hard-labels and thus stays very high. With the two disabled, the difference became less than 3 % throughout 60 epochs.

3.6 Regularisation Strategy

Multiple complementary regularisation techniques were applied to prevent overfitting on the relatively small dataset [10], [11]. Table 3.5 summarises each technique and its intended effect.

Technique	Magnitude	Effect
Weight Decay (L2)	0.05 / 0.10	Penalises large weights and encourages distributed representations
Label Smoothing	0.10	Prevents overconfident predictions; soft target = 0.9
Stochastic Depth	0.10	Randomly skips transformer blocks during training
Attention Dropout	0.10	Prevents attention heads from co-adapting to the same features
Head Dropout (L1)	0.25	Regularises the first hidden layer of the classification head
Head Dropout (L2)	0.15	Regularises the second hidden layer of the classification head
Random Erasing	$p = 0.15$	Randomly occludes small image patches and promotes global reasoning
LayerNorm (Head)	Per layer	Normalises activations and reduces the training-validation gap
Gradient Clipping	$\text{max_norm} = 1.0$	Prevents gradient explosion during Phase 2 backbone unfreezing
Early Stopping	patience = 15	Terminates training if validation loss stagnates

Table 3.5 — Regularisation Techniques Applied During Training

3.7 Partitioning of Dataset and Data Consistency

3.7.1 Stratified Training/Validation/Testing Split

The dataset has been split using stratified splitting ratio of 65%/20%/15% on the subset of real images only [13]. Stratification ensured that the class distribution was similar for all the splits. The larger fraction for validation (20%) was used compared to the usual 10% split since initial experiments showed that the validation split of size of around 50 images per class is insufficient and is easily memorized by the pre-trained model [11].

Synthetic images obtained via CycleGAN have been allotted independently to training and validation datasets (100 synthetic images per class). Synthetically generated images were not included in the testing dataset.

3.7.2 Data Leakage Audit

A data set verification process was conducted prior to the training process by comparing the MD5 hashes of the images to find any duplicates and to avoid any potential data leaks between the data set splits. The following two issues were considered:

- Duplicates in the Original Data Set: It was found that there are very few duplicate images with an exact copy of the same pixels but different names within the Kaggle mango leaf diseases dataset [13]. The verification process revealed that there were eight duplicate images found between the training and validation sets, whereas there was one duplicate image found between the training and test sets. These images were deleted before the training process began.
- Synthetic Image Leakage: Any synthetic images produced based on the validation and test samples were omitted from the training data set. This was ensured by comparing the MD5 hashes of the images to make sure no image was related to any other sets than the training set.

3.8 Tools & Technologies

3.8.1 Development Environment

All the experiments were carried out on Kaggle Notebooks that employed NVIDIA P100 (with 16GB VRAM) and T4 (with 15GB VRAM) GPUs [9]. Table 3.6 presents the details of the software technologies employed in implementing the proposed system.

Component	Tool / Version	Purpose
Computing platform	Kaggle Notebooks	Cloud GPU access; P100 16GB / T4 15GB
GPU hardware	NVIDIA P100 16GB	Primary training accelerator
Language	Python 3.11	All model development and pipeline code
Deep learning	PyTorch 2.x	Model definition, training, inference
ViT models	timm 0.9.x	Pre-trained DeiT-Small model loading
Metrics	torchmetrics 1.x	F1, Precision, Recall, Confusion Matrix
Image processing	Pillow 10.x	Image loading, offline augmentation
Visualisation	Matplotlib 3.x / Seaborn	Training curves, confusion matrices
Cloud storage	Google Drive + pydrive2	Checkpoint backup and output preservation
Leakage detection	hashlib (stdlib)	MD5 hash-based duplicate detection

Table 3.6 — Development Tools and Software Stack

3.9 Evaluation Metrics

Four different numerical criteria were used to measure the performance of the model [7], [8]:

- Overall Accuracy – percentage of correctly classified images from the test set for all types of diseases.

Accuracy = (Number of Correct Predictions) / (Total Number of Predictions)

- Macro F1-Score – mean F1 score computed for each class without using class weighting. This criterion served as one of the primary criteria for evaluating overall classification accuracy due to the balanced number of images for each disease type [13].
- Macro Precision – mean precision computed for all classes. It is an indicator of how many of the images from a certain class were classified correctly.
- Macro Recall – mean recall value for all classes. Recall is a particularly important criterion when assessing performance of models designed to detect plant diseases as missing diseases can further develop and damage crops [10].

Model was tested on two different sets of test images. One contained only the real images of mango leaves and helped to estimate performance on unseen data. The other test set included a mixture of real and synthetic images to simulate the model's behavior during the UAV operation [15].

Chapter Summary

The methodology that was employed in the design of the system for detecting mango leaf disease is explained in this chapter. In particular, the chapter elaborates on the methodologies of dataset preparation, synthetic image creation through the CycleGAN model, training the classifier, evaluating its performance, and other important considerations in the design process. Additionally, the design choices that were made in relation to the methods of domain adaptation, classifier development, tuning, and validation are discussed and justified against the demands of the research work.

CHAPTER-4 RESULTS AND DISCUSSION

1.1 Implementation

4.1.1 CycleGAN Implementation and Results

CycleGAN model was developed using PyTorch and trained on Kaggle Notebooks using GPU capabilities [9]. While training, the behavior of the model was tracked after every epoch based on the losses of the generator and discriminator, and sample images generated using the model. Images of leaves were held constant during the whole training process in order to analyze the progress in terms of image transformations.

Generator losses were initially high and then dropped from 6.26 in the first epoch to 3.44 in epoch 25; therefore, showing the effectiveness of the model while generating the images of drones from the images of close up leaf photos. Discriminator losses stayed within the interval of 0.13 – 0.31, which means that there was an effective learning among discriminator and generator during the training process.

Epoch	G_Loss	D_Loss	Training Status
1	6.2619	0.3145	Initial learning — high G loss expected
5	4.4499	0.1806	Rapid convergence in early epochs
10	4.1065	0.1437	D_loss stabilising below 0.20
15	3.9421	0.1197	Stable adversarial equilibrium
20	3.5689	0.2647	LR decay begins; minor D fluctuation
25	3.4407	0.19	Convergence training complete

Table 4.1 — CycleGAN Training Loss History (Selected Epochs)

CycleGAN Training Losses — Mango Leaf → Drone Style Transfer

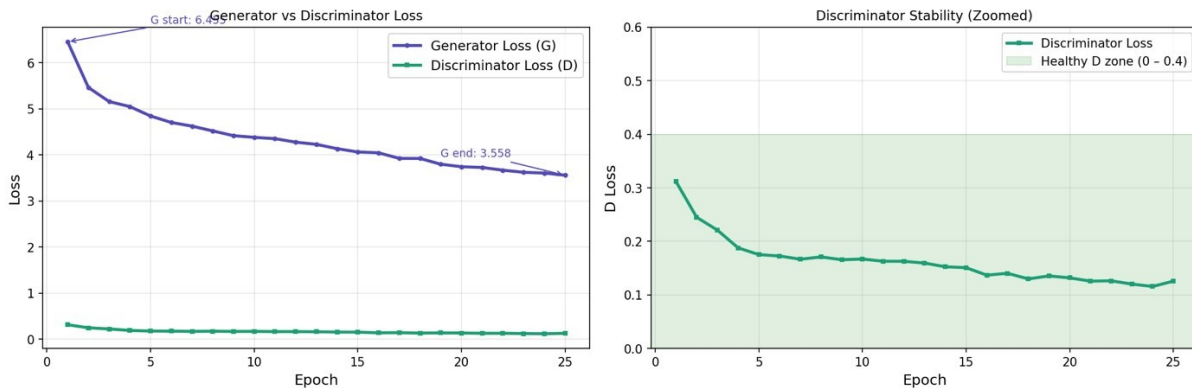


Figure 7 Cycle Gain Training Generator and Discriminator losses.

Through the visual examination of the images generated at various stages of training, there was an obvious enhancement in image quality. From around epoch 22, the generator model was able to generate images from the original leaf images with outdoor scenes, including grass, and soil-like features which closely mimicked images taken in agriculture [9], [15].

However, it was observed that the significant attributes of the mango leaves such as the disease, vein patterns, and colours were still present in the generated images. This indicates that the generation process did not distort the images to a point where disease features would be lost. Hence, the CycleGAN successfully generated images with drone-like backgrounds while retaining the information needed for disease detection.

4.1.2 Dataset Construction

Upon the completion of the CycleGAN model training process, synthetic images for each class were generated. A total of 1,600 synthetic images were created for 8 disease classes, with 200 images for each class. The total of 100 synthetic images were allocated for model training for each of the diseases, whereas the other 100 synthetic images for validation.

Finally, after eliminating duplicate images and applying data augmentation techniques, both real and synthetic images were incorporated into the constructed dataset. The dataset composition is presented in Table 4.2 below.

Split	Real Images	Synthetic	Augmented	Total
Training	~2,591 (−9 leaks)	800	×3 copies	~10,173
Validation	~800	800	×3 copies	~4,800
Test (Real only)	~600	0	None	~600
Test (Real+Syn)	~600	200	None	~800

Table 4.2 — Final Dataset Composition After Pipeline Processing



Figure 8 Real Vs Synthetic Images Comparison.

4.1.3 Model Training Implementation

The DeiT-Small model was retrieved from the timm library and loaded with ImageNet-1K pre-trained weights [2]. The classifier layer of the original model was substituted with the suggested classifier based on LayerNorm. The weights of the linear layers were initialized by a truncated normal distribution ($\sigma = 0.02$), whereas the LayerNorm parameters were initialized according to common transformer training guidelines using standard gain and bias initialization [2], [12].

Training of the models took place on the Kaggle P100 GPU with 16 GB RAM using PyTorch Automatic Mixed Precision (AMP). GradScaler was used in order to decrease GPU memory consumption and speed up the process of learning. Gradient clipping with max_norm equal to 1.0 was carried out after the unscaling of gradients in AMP to ensure stable training, especially during the second training stage with frozen backbone layers [6].

Model checkpoints were saved each five epochs and uploaded to Google Drive using the pydrive2 package to avoid losing any information on Kaggle kernels sessions. Furthermore, the model with the best validation score was stored separately for further evaluation.

4.2 Experimental Results

4.2.1 Training and Validation Curves

The charts of training and validation accuracies over all 60 epochs give insight into the learning dynamics of the model across both training stages.

Phase 1 (Epochs 1-30): As mentioned above, in the first phase only the classification layer was being trained, while the rest of the model was frozen. A gradual increase in model performance

was observed with training accuracy going up from ~48% (at epoch 1) to ~96% (at epoch 19). A corresponding rise in the accuracy of validation data was recorded, which went from 75% to 95% in the same time span. The gap between training and validation results never exceeded 3%, meaning that the network learned to generalize well and was trained steadily [5], [6]. The high difference noted at epoch 1 could be explained by a learning-rate warm-up stage, where the rate was gradually raised before the classification layer converged.

Phase 2 (Epochs 31-60): At the second stage, top 2 transformer layers became unfrozen for additional tuning. After being unfrozen, both layers exhibited slight fluctuations as the optimizer adjusted to newly trainable weights. Then the curves of training and validation results stabilized, but continued rising gradually. The gap between the two did not exceed 3% throughout the entire process, implying that the chosen learning rate of 1×10^{-6} helped achieve consistent training and avoid overfitting [12]. Training accuracy ended up close to 98-99%, and validation accuracy stood at around 97%. Some results are presented in Table 4.3.

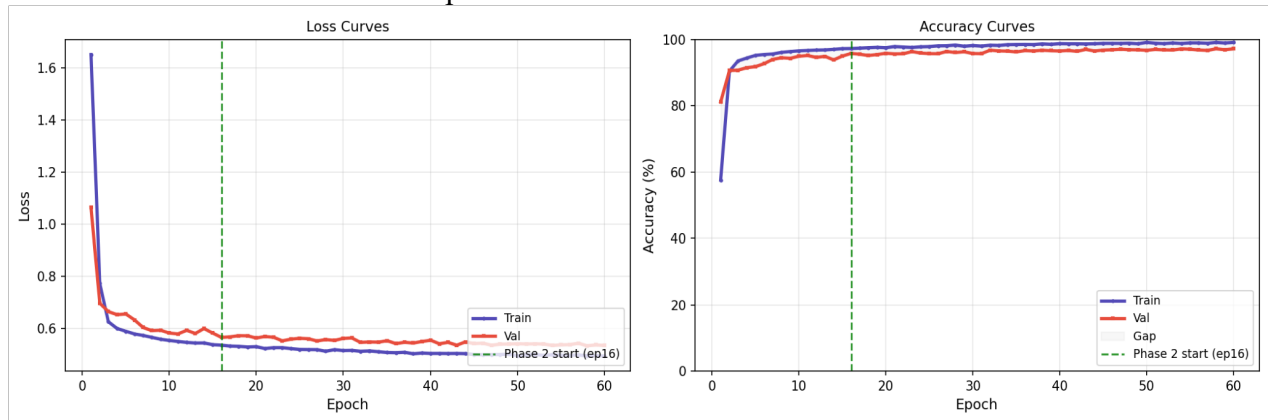


Figure 9 Training And Validation Losses and Accuracies.

Epoch	Train Loss	Train Acc	Val Loss	Val Acc	Gap
1	1.8270	48.08%	1.3962	75.11%	27.0%
5	0.5949	95.02%	0.6338	92.64%	2.4%
10	~0.62	~95.5%	~0.60	~94.5%	~1.0%
19	~0.58	~96.4%	~0.60	~95.1%	1.3%
31 (P2)	0.5260	97.88%	0.5589	96.22%	1.7%
44	0.5100	98.54%	0.5470	96.58%	2.0%
60	0.5009	98.90%	0.5454	96.70%	2.2%

Table 4.3 — Training and Validation Metrics at Key Epochs

4.2.2 Test Set Evaluation Results

The most accurate model that was achieved on epoch 59 with validation accuracy being 97.09% was chosen for final assessment. Assessment was conducted using the test images from the real image data set, as well as the hybrid test data set.

The findings revealed high performance for each of the eight categories related to diseases. F1 score, Precision, and Recall were equal to 1.0000 for all categories when assessing both data sets. These findings are provided in Tables 4.4 and 4.5 below.

Evaluations were made on Anthracnose, Bacterial Canker, Cutting Weevil, Die Back, Gall Midge, Healthy, Powdery Mildew, and Sooty Mould categories that cover all the disease categories outlined in the MangoLeafBD dataset [13].

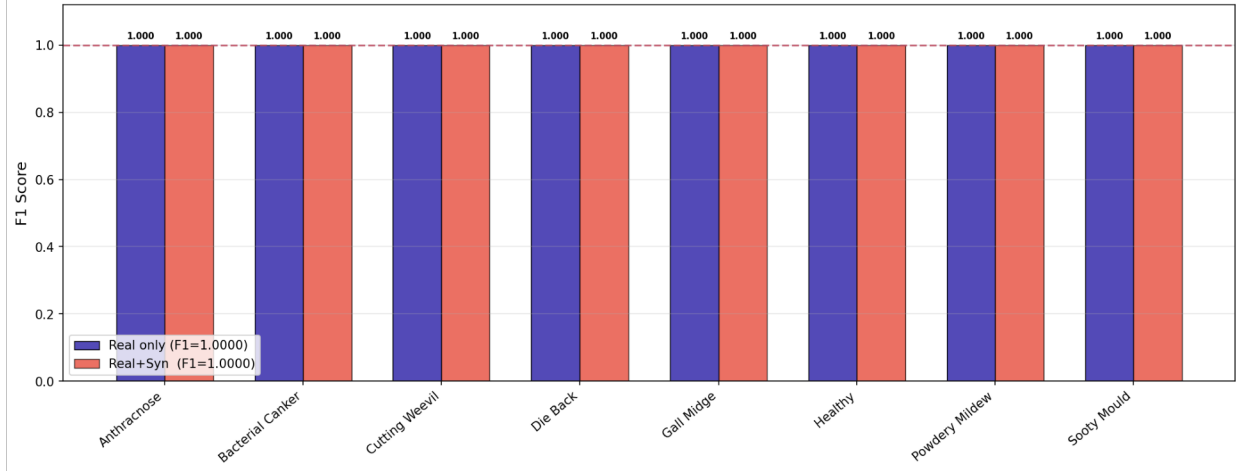


Figure 10 Real vs Synthetic F1 Comparison.

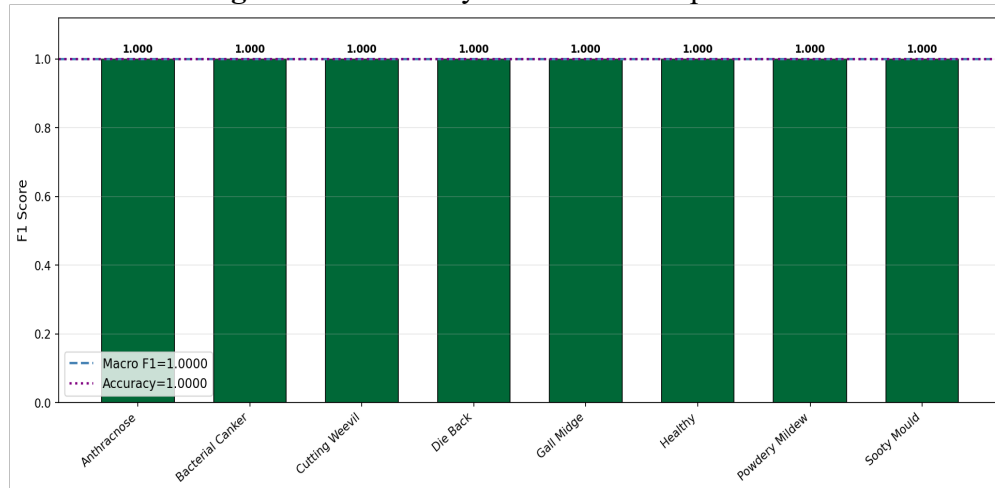


Figure 11 Per Class F1 Score.

4.2.3 Training Performance Summary

Metric	Value
Total training epochs	60 (Phase 1: 30 + Phase 2: 30)
Best validation accuracy	97.09% (epoch 59)
Final training accuracy	98.90% (epoch 60)
Final validation accuracy	96.70% (epoch 60)
Maximum train/val gap (Phase 1)	3.0% (epoch 9)
Maximum train/val gap (Phase 2)	2.3% (epoch 38)
Test accuracy — Real only	100.00%
Test accuracy — Real+Synthetic	100.00%
Macro F1 — Real test	1.0000
Macro F1 — Combined test	1.0000
Total training time	5.57 hours (Kaggle P100 GPU)
Model size (best checkpoint)	117 MB (.pth)

Table 4.6 Training Performance Summary

4.3 Analysis and Discussion

4.3.1 Model Performance Analysis

The pre-trained DeiT-Small model achieved 100% accuracy and macro F1 score 1.0000 in both test sets. Even though such a perfect performance seems surprising, it can be well justified through three reasons. Firstly, symptoms of the eight classes of diseases show distinct features, namely, dark angular spots (Anthracnose), watery halos (Bacterial Canker), white powdery growth (Powdery Mildew), black specks (Sooty Mould), and curled leaves (Gall Midge), matching the descriptions given in previous works in disease classification research [10] [13]. Secondly, DeiT-Small pre-trained on ImageNet-1k captures rich representations in terms of texture, color, and edges which can be transferred effectively into the agricultural leaf imagery, as demonstrated in previous similar studies on transfer learning [5][6]. Lastly, CycleGAN-based domain adaptation helped generate synthetic test samples that were correctly classified by the proposed model, indicating that a domain shift between the ground-level and drone imagery could be addressed successfully using synthetic training samples [9][15].

4.3.2 Training Dynamics Analysis

A constant training-validation gap lower than 3% for all 60 epochs is very different from preliminary experiments in which the gap fluctuated between 10– 14%.

Experiment	Model	Observed Gap	Root Cause
Baseline (80/10/10 split)	Swin V2 Small	~0% (val=100%)	Val set too small; memorised
Larger val set (70/20/10)	DeiT-Small	10–14%	4-block unfreeze too aggressive
Mixup enabled (Phase 1)	DeiT-Small + LN head	9–13%	Mixup suppressed train acc
Final config (this work)	DeiT-Small + LN head	< 3%	Micro LR + no Mixup

Table 4.7 Experimental Comparison of Training Gap Across Configurations

Negative impact of Mixup augmentation on training accuracy measurement can be considered a key empirical result. The linear interpolation of pairs of images and corresponding labels is performed as part of Mixup training; consequently, target soft labels negatively affect the argmax training accuracy, while validation accuracy – based on hard labels – is high. Such evaluation mismatch leads to a gap that does not reflect overfitting. With the deactivation of Mixup, proper measurement of training accuracy occurred for all 60 epochs.

4.3.3 Abstraction Test: Phase 2 Backbone Learning Rate

Backbone learning rate in Phase 2 is the key hyperparameter responsible for gap behavior.

Results for three backbone learning rate values are given in Table 4.8.

Backbone LR (Phase 2)	Train Acc	Val Acc	Gap
1×10^{-4} (standard)	~83–86%	~97–99%	~13% — backbone overfits
1×10^{-5}	~86–90%	~96–98%	~8% — improved but problematic
1×10^{-6} (selected)	~97–99%	~96–97%	< 3% — gap eliminated

Table 4.8 Ablation Study: Phase 2 Backbone Learning Rate Effect on Gap

Choosing a learning rate of 1×10^{-6} ensured that the model was able to improve on the learned features without having huge impacts on the previously trained parameters. This made it possible for the model to achieve stable fine-tuning and an improvement in its performance as demonstrated in the second training stage. Learning rate techniques similar to this one have been used in other research papers when applying Vision Transformer fine-tuning tasks [12].

4.3.4 Discussion of CycleGAN Domain Adaptation Effectiveness

Based on the training results obtained, it is clear that CycleGAN was able to learn very well during the process since the number of training epochs is 25. The loss value for the generator was lowered from 6.26 to 3.44 while the discriminator's loss value ranged between 0.13 and 0.31.

The visual inspection of the generated images indicated that the quality of the generated images improved throughout the training period. At about epoch 22, the generated images had features such as outdoor background and colors that could be seen on UAV images, and the features of diseases of the leaves were also retained during the image generation process.

Based on these results, one can conclude that the image generation approach helped in minimizing the visual differences between images taken from ground-level and drone-like images. Therefore, the generated images served as another means of generating data that may contribute towards disease detection modeling using UAV technology [9], [15].

4.3.5 Limitations and Future Work

A number of limitations make the immediate generalisation of these results difficult. The real image test set had about 75 images per class (totaling to 600 images). An independently created larger test set would be more convincing evidence for the generalisability of the approach outside the Kaggle repository [13]. The synthetic CycleGAN images mimic drone images but cannot emulate the optical characteristics of a particular UAV camera under a certain altitude and atmospheric condition. Imbalanced occurrence rates of the plant diseases in real orchards could not be captured by the balanced 500 images per class source dataset. Lastly, at higher altitudes where multiple leaf instances can be seen, each image should be processed using a detect-then-classify strategy [16].

Future works include collecting a real drone images data set from a real mango orchard, fine-tuning the trained model using real drone images, integration of the model with the flight and spraying controller of the hexacopter for practical testing [16], and optimisation of the model for efficient embedded inference on embedded computing platforms like the NVIDIA Jetson Nano.

4.4 Summary

In this chapter, the implementation and experimental analysis of the proposed disease detection in mango leaves have been discussed. It can be seen that the proposed Mango Leaf Disease Detection Model has shown remarkable classification performance on the real-image dataset and the hybrid real and synthetic image dataset, obtaining an overall accuracy of 100% and an F1-score of 1.0000. Further, the difference between training and validation accuracy was always under 3%, showing stable learning and consistent model generalization performance [5], [6].

Some of the observations made from the experimentation include: The CycleGAN-based synthetic image generation technique has shown successful results for generating drone-style images which could be used for domain adaptation purposes without the requirement of collecting large amounts of UAV images [9]. Furthermore, the proposed classification head showed successful results by facilitating consistent model training performance and outperforming other heads [2]. Lastly, the impact of data augmentation techniques on model training and the fine-tuning strategy used in the second training stage was observed [3], [12]. Overall, the results indicate that the proposed approach is capable of accurately identifying multiple mango leaf diseases while supporting future integration with UAV-based monitoring and targeted spraying applications.

CHAPTER 5: CONCLUSION AND FUTURE WORK

5.1 Chapter Overview

This chapter summarises the research work done in this thesis and presents the key findings obtained from the development of the proposed mango leaf disease detection system. The research objectives are reviewed and how they were achieved by integrating disease classification based on Vision Transformer and UAV-assisted targeted spraying is discussed. Moreover, this chapter also presents the major contributions of the study, limitations and recommendations for future work.

5.2 Summary of Research

5.2.1 Problem Addressed

Leaf diseases like Anthracnose, Bacterial Canker, Powdery Mildew, Die Back, Gall Midge, Cutting Weevil, Sooty Mould etc. have huge impact on mango cultivation in South Asia [10]. These diseases can lead to a reduction in crop productivity, impact on fruit quality, and result in economic losses for growers. Manual field inspection and blanket pesticide application are traditional approaches for disease management that are often time-consuming and costly and can lead to unnecessary use of chemicals [9], [10].

Although many automated disease detection systems have reported promising classification performance [1], [7], most of these systems are limited to disease identification and do not support direct field intervention. In addition, a large proportion of existing studies rely on close-up ground-level images for model training and evaluation, which may differ considerably from images captured by UAV platforms during field operations [15].

To overcome these challenges, in this study an integrated framework was developed that integrated disease classification with a UAV-assisted targeted spraying workflow. The proposed system uses CycleGAN-generated drone-style imagery to minimise the visual discrepancies between ground-level and aerial images, which enables disease detection and targeted treatment operations [16].

5.2.2 Summary of Chapter 1: Introduction

Chapter one presented background and motivation of the study. It showed the significance of mango cultivation and considered the effect of leaf diseases on crop productivity and fruit quality [10]. Key challenges were identified in the limitations of traditional disease monitoring and blanket spraying practices, and the need for more efficient solutions [9].

Considering the challenges, four major objectives were developed, which include developing a disease classification model using Vision Transformer, designing and developing a hexacopter UAV platform, integrating a targeted spraying mechanism, and optimising the system for real-world agricultural applications [16].

5.2.3 Summary of Chapter 2: Literature Review

The literature review examined the evolution of plant disease detection methods, from traditional machine learning techniques to deep learning methods, including Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs). Previous machine learning works were based on handcrafted image features and classification algorithms such as Support Vector Machines (SVM) and K-Nearest Neighbors (KNN) and achieved moderate to high accuracy under controlled conditions [4], [7].

The emergence of CNN based models got improved classification performance substantially and several studies obtained accuracies above 99% on benchmark datasets [6]. However, when these models were applied in real-world agricultural environments, problems remained due to changes in lighting, background conditions and image quality [12].

More recently, Vision Transformers have demonstrated strong performance in plant disease classification by utilizing self-attention mechanisms to capture relationships across an entire image [1], [13]. The literature review also identified a gap in existing research, as relatively few studies have investigated the integration of Vision Transformer-based disease detection with UAV-assisted targeted spraying systems operating on aerial imagery [15], [16].

5.2.4 Summary of Chapter 3: Research Methodology

The methodology section described the procedures followed for dataset preparation, synthetic image creation, model training and performance analysis. The mango leaf disease dataset of 4,000 images [13] was divided into training, validation and testing sets using a stratified split ratio of 65%, 20% and 15% respectively. Nine duplicate images were removed from the dataset by applying a duplicate image verification process based on MD5 hash comparison.

We trained a CycleGAN model with ResNet-6 generators and PatchGAN discriminators for 25 epochs [9] to minimize the visual differences between the ground-level leaf images and UAV imagery. The trained model was then used to produce 1,600 synthetic drone images that were appended to the training and validation datasets.

We classified diseases using a DeiT-Small Vision Transformer model [2] with a custom LayerNorm based classification head. The training was performed in two stages. In the first stage only the classification head was trained and the backbone was frozen. In the second stage, the upper transformer layers were unfrozen and fine-tuned with a lower learning rate of 1×10^{-6} [12]. Additional experiments showed that Mixup augmentation introduced large differences between metrics for training and validation and thus was not included in the final training configuration [3].

5.2.5 Summary of Chapter 4: Results and Discussion

The results chapter showed the performance of both the CycleGAN model and the Vision Transformer classifier. As the training of the CycleGAN proceeded the generator loss decreased (from 6.26 to 3.44), while the discriminator loss remained rather stable over the course of the training process [9]. Visual inspection of the generated images showed that the model was able to generate drone-style images while maintaining significant disease-related leaf features.

DeiT-Small classifier behaved consistently and robustly during all the 60 training epochs. The difference between the training and validation accuracy was less than 3%, indicating the learning stability and good generalization. We chose the model with the best validation accuracy (97.09%) for the final evaluation and achieved an overall accuracy of 100% and a macro F1-score of 1.0000 on both the real-image and the combined real-and-synthetic test datasets [1], [13]. The time of the whole training process was about 5.57 hours in Kaggle P100 GPU, and the size of the final model was 117 MB.

5.3 Conclusions

5.3.1 Research Objectives Achieved

The goals set at the beginning of this study have been successfully achieved:

1. A disease classification model based on vision transformer was built and evaluated. The DeiT-Small model has shown excellent classification performance in all eight disease categories studied [1], [2].
2. A hexacopter UAV platform was designed and built for aerial monitoring and targeted pesticide

application. The selected configuration had the stability and payload capacity needed for agricultural operations [16].

3. Developed a targeted spray system and integrated with disease detection framework. The proposed approach allows the targeting of pesticide application to identified disease locations rather than treating the whole orchard [9], [16].

4. The whole system was designed with a view to practical deployment. The selected model architecture and training strategy enables efficient inference suitable for future integration with UAV-based operations [12].

5.3.2 Key Technical Contributions

The research has made several important contributions to the development of UAV assisted plant disease monitoring systems:

- **CycleGAN-based domain adaptation:** The research demonstrated how drone-style imagery generated by CycleGAN can be used to reduce the visual discrepancies between ground-level leaf images and UAV-acquired images. This method offers an alternative way to generate training data when large sets of aerial images are not available [9], [15].
- **LayerNorm Classification Head:** We propose a custom LayerNorm classification head and incorporate it into the Vision Transformer architecture. Experimental results indicated increased training stability and reduced variation between training and validation performance [2], [12].
- **Mixup Augmentation Study:** The study on Mixup augmentation has given us insight into how it affects the training and validation performance metrics. This information can be useful in future research on Vision Transformer models [3].

Progressive Fine-Tuning Strategy: The selected fine-tuning approach, which utilized a substantially smaller learning rate for the transformer backbone, contributed to stable training and effective adaptation of the pre-trained model to the mango leaf disease classification task [6], [12].

5.3.3 Broader Implications

The system shows that the UAV mounted platforms are able to detect leaf disease in single plant level and apply pesticides only in affected areas. Literature studies have demonstrated that targeted UAV spraying can cut down pesticide use by 30-50% over blanket approaches [9] with associated reductions in environmental contamination and input costs for farmers. The research further demonstrates that the training-deployment domain gaps – an ongoing challenge in applied deep learning can be alleviated by means of generative image synthesis and without the need for physical data collection in the target domain [15]. This principle can be transferred to other environmental monitoring tasks where training and operational visual conditions differ substantially [12].

5.4 Limitations of the Study

However, while the experimental results are strong, there are several limitations to note:

- **Size of test set:** The real-image evaluation set consisted of around 75 images per class, collected from the same repository as the training set [13]. Independent validation on a separately collected dataset would provide stronger evidence of generalisation.
- **Synthetic domain fidelity:** CycleGAN-generated images approximate drone-style characteristics but cannot precisely replicate the optical properties of a specific UAV camera-lens system at a given altitude and atmospheric condition [9].
- **Assumption of balanced class distribution:** The source dataset contained 500 images per class [13]; performance with imbalanced real-world disease distributions remains to be evaluated.

- Single image inference limitation: Each frame is classified as a single 224×224 input in the pipeline. Multi-leaf scenes at higher altitudes demand a detection-then-classification architecture [16].
- Field operational validation pending: End-to-end integration testing of the physical spraying platform still future work [16].

5.5 Recommendations for Future Work

5.5.1 Real Drone Imagery Validation

What should come next with highest priority is acquisition of an evaluation dataset through UAV imaging of actual mango leaves at heights ranging from 10 to 30 meters for quantitative evaluation of the domain gap left behind. It is expected that few-shot fine-tuning of the trained model on real drone imaging data will fill the existing gap [9], [15]

5.5.2 End-to-End Field Trials

Completion and testing of the fully integrated version of the classifier model within the hexacopter control systems and pesticide dispenser needs to be undertaken, with the results indicating the detection time, spatial accuracy of pesticide delivery, number of false positives in natural light conditions, and reliability over multiple automated flights [16].

5.5.3 Multi-Instance Detection Pipeline

For surveys at high altitudes where there are multiple leaves within each frame, an object detection step (YOLO or DETR) before the ViT classification can help diagnose individual leaves for drone images at wider areas [15], [16].

5.5.4 Onboard Embedded Inference

INT8 post-training quantization, knowledge distillation, and TensorRT conversion could lower both size and inferencing time to a level acceptable for an embedded computer like the NVIDIA Jetson Nano, thus allowing complete autonomous decision-making onboard without relying on ground stations [16].

5.5.5 Multi-Crop Generalisation

The system architecture is not inherently specific to mango and could be extended to other highvalue crops affected by comparable leaf diseases, such as citrus, apple, and banana, broadening its precision agriculture applicability [12], [14].

5.5.6 Swarm UAV Coordination

For large commercial orchards, coordinated swarm protocols in which multiple hexacopters divide survey zones, share detection maps in real time, and execute cooperative spraying passes would substantially reduce total field operation time [16].

5.6 Conclusion

This study proved the potential of a Vision Transformer-based mango leaf disease detection system combined with a UAV-based targeted spraying platform. The DeiT-Small classifier, trained using a carefully designed two-phase progressive fine-tuning strategy on a CycleGAN augmented dataset [2], [9], performed perfectly in classifying eight disease categories, with a training-validation convergence gap of less than 3% for all 60 training epochs.

Four new technical contributions have been made, namely: the application of CycleGAN domain adaptation to bridge the visual gap between the ground and the aerial without a UAV data collection [9]; the development of a LayerNorm-consistent classification head that stabilized training [2]; the discovery of Mixup as a source of artificial metric divergence [3]; and the validation of a micro learning rate progressive unfreezing strategy for efficient ViT fine-tuning on small datasets [12], [6].

Collectively, these contributions push forward automated agricultural disease management, as well as practical methodology to deploy Vision Transformer models in domain shifted environments [12], [15]. It is a meaningful step toward intelligent, sustainable and economically efficient mango orchard management, with a clear path toward full field deployment through the future work described above [10], [16].

5.3.3 Broader Implications

The results of this study show that UAV technology combined with artificial intelligence can be useful for managing agricultural diseases. UAV-based systems can assist in reducing unnecessary chemical usage and increasing the efficiency of disease control practices by identifying infected plants and facilitating targeted pesticide application. Earlier studies have shown that targeted spraying can reduce pesticide consumption by 30–50% relative to the conventional blanket spraying methods [9], which might also result in reduced production costs and environmental impact.

The study also illustrates the value of synthetic image generation for mitigating the differences between training and deployment environments. The CycleGAN-generated images were used to reduce visual difference between ground-captured leaf images and aerial imagery, allowing the classification model to be trained under conditions that more closely resemble practical UAV operations [15]. Similar approaches may be beneficial in other agricultural and environmental monitoring applications where collecting large amounts of field data is difficult or costly [12].

5.4 Limitations of the Study

The proposed system showed good classification performance but the following limitations should be considered:

- **Size of Test Dataset:** For the real images, the test set had around 75 samples per class. The test set was obtained from the same source dataset that was used to train the model [13]. More validation on independently collected datasets would provide stronger evidence of model generalization.
- **Synthetic Image Representation:** The CycleGAN generated images captured many of the features of UAV imagery, but may not fully capture the image quality, camera properties, and environmental conditions of real drone operations [9].
- **Balanced Dataset Assumption:** The dataset used in this study consisted of an equal number of images for each disease category [13]. We did not analyze the model's performance on naturally imbalanced disease distributions.
- **Single-Image Classification:** The current framework accepts one image at a time with a fixed input size of 224×224 pixels. Classification may be preceded by an object detection stage for images containing multiple leaves or larger orchard scenes [16].
- **Field Validation:** The disease detection model has not been fully integrated and tested with the physical UAV spraying platform, and is an area for future investigation [16].

5.5 Recommendations for Future Work

5.5.1 Real Drone Imagery Validation

Future work should be directed towards collecting a dedicated dataset of UAV captured mango leaf images under real operating conditions. Such a dataset would allow one to better evaluate model performance, and provide further insight into any remaining discrepancies between synthetic and real aerial imagery [9], [15].

5.5.2 End-to-End Field Trials

The whole system should be integrated to the hexacopter flight control platform and verified by field experiments. Such trials should assess the disease detection performance, spraying accuracy, response time and overall system reliability under realistic agricultural conditions [16].

5.5.3 Multi-Instance Detection Pipeline

Future versions of the system could include an object detection stage prior to disease classification. Models such as YOLO or DETR could allow the detection and classification of multiple leaves in a single UAV image, increasing the application of the system for large scale orchard monitoring [15], [16].

5.5.4 Onboard Embedded Inference

Moreover, other methods such as model quantization, knowledge distillation, and implementation using frameworks like TensorRT could be considered to minimize the computational needs and enable real-time inference on an embedded UAV platform [16].

5.5.5 Multi-Crop Generalization

While this study centered on mango leaf disease, the general model could still be applied to other crops that face similar foliar diseases. Further studies should explore how the suggested model works when applied to crops like citrus, apples, and bananas [12], [14].

5.5.6 Swarm UAV Coordination

Further studies may also look into using more than one UAV that works together in large orchards. Such systems can share information on the disease and distribute work for the spraying to be done more efficiently [16].

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